UB PHC 607/608/609 - Clinical PK/PD as Dose Individualization

Based on Guohua An's Buffalo notebooks, rewritten as an original study text

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Clinical pharmacokinetics and pharmacodynamics 2026

Reading motive
这份笔记把 Buffalo PHC 607/608/609 相关材料重写成一卷完整的自学文本。Guohua An 的两本 notebook 给出课程的覆盖面：clinical pharmacokinetics, dose optimization, biologics, advanced PK/PD, PBPK, population PK, NONMEM, pharmacodynamics, indirect response, and QSP [2, 1]. 这里不做逐句改写，也不做替代教材式复制；我把它们当作 topic map，然后用我们前面形成的 dec41/pdf2htmlEX 风格写成一条原创推理线。 [source map]

Audience
读者可以没有系统学过 PK/PD。全书从 dose, concentration, amount, clearance, volume, probability, and decision 开始，然后逐步走到 clinical dose adjustment, biologics, nonlinear kinetics, population PK, PBPK, PD, TMDD and QSP. 目标不是背参数，而是让读者知道一个临床剂量问题如何变成可计算、可解释、可质疑、可更新的模型问题。 [from zero]

Style
中英文混排。中文负责逻辑铺垫和判断，英文保留药动药效、模型和软件语境里最自然的术语。符号沿用我们之前的规则：random variables 用 sans-serif，Greek random quantities 用 bold Greek，多字母 PK/PD 固定搭配在数学公式中用 text italic label，比如 $CL, AUC, PK, PD, PBPK, QSP$。 [notation]

Contents

1 The Unifying Problem

1.1 Dose is not the object; decision is the object

Clinical pharmacokinetics 常常从一句话开始：what the body does to the drug. 这句话没错，但它太静态。真正的课程主线应该是：

$$\text{dose} \longrightarrow \text{exposure} \longrightarrow \text{response} \longrightarrow \text{decision} \longrightarrow \text{next dose.}$$

If dose were the object, the course would end after prescription. If exposure were the object, it would end after plotting concentration–time curves. But clinical PK/PD is a decision science: we ask how a dose should be chosen, adapted, delayed, held, escalated, de-escalated, or individualized under uncertainty.

This is why PHC 607/608/609 naturally form one book. PHC 607 names the objects: concentration, amount, absorption, distribution, clearance, bioavailability, half-life, and exposure. PHC 608 turns those objects into dosing operations: loading dose, maintenance dose, infusion, accumulation, fluctuation, nonlinear elimination, special populations, therapeutic drug monitoring. PHC 609 then asks what happens when the simple objects are no longer sufficient: advanced PK/PD, PBPK, population models, biologics, target-mediated disposition, indirect response, and QSP.

1.2 The central equation

The most compact form of the whole course is

$$p_\theta(y_i, \psi_i, r_i \mid u_i, c_i) = p_\theta(r_i \mid y_i, \psi_i, u_i, c_i)\, p_\theta(y_i \mid \psi_i, u_i)\, p_\theta(\psi_i \mid c_i).$$

Here u_i is the dosing and sampling design, c_i is covariate information, ψ_i is individual physiology and pharmacology, y_i is observed data, and r_i is response or clinical outcome. The first semester asks how u_i becomes y_i. The second asks how to use y_i to choose the next u_i. The third asks how to learn p_θ from many patients and then use it responsibly.

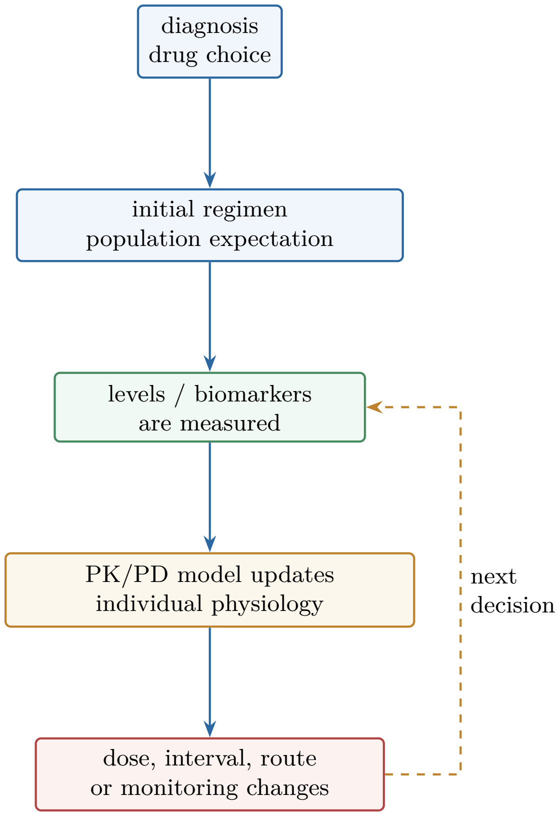


Figure 1: A clinical PK/PD decision loop. This is an original redraw of the TDM workflow idea: diagnosis starts therapy, data update the model, and the next regimen is a decision under uncertainty.

Remark. Classical PK looks deterministic because many textbook equations are deterministic. Clinical PK is not deterministic. The deterministic equation is the forward map inside a larger uncertain system: uncertain patient, uncertain adherence, uncertain sampling time, uncertain assay, uncertain response, uncertain future.

1.3 What makes a clinical PK/PD mind

A clinical PK/PD mind learns to ask five questions before touching software:

(i) What is the decision? Dose, schedule, label, trial design, mechanism, or prediction?

(ii) What is observed? Concentration, urine amount, biomarker, response, toxicity, survival, or only sparse clinical data?

(iii) What is latent? Individual CL, V, target expression, enzyme activity, adherence, effect-site concentration, or disease state?

(iv) What is shared across patients? Typical values, variability, covariate effects, residual error, physiological constants, or mechanistic pathways?

(v) What would falsify the working model? Wrong slope, wrong peak, wrong trough, wrong variability, wrong delay, wrong dose-response, or wrong special-population prediction?

The answer to these questions decides whether a one-compartment model is enough, whether NCA is the right summary, whether a well-stirred liver model is appropriate, whether a biologic needs TMDD, whether NONMEM is necessary, and whether a QSP model is credible.

2 Notation and Symbol Discipline

2.1 Random variables and realized values

We adopt the inference notation used in the earlier Bayesian PopPK note. A random concentration before observation is $\mathsf{C}(t)$; an observed concentration is $C(t)$. A random individual clearance is CL_i; a realized or candidate value is CL_i. Greek random quantities use bold Greek:

$$\boldsymbol{\eta}_i \sim \mathcal{N}(0, \Omega), \qquad \boldsymbol{\epsilon}_{ij} \sim \mathcal{N}(0, \sigma^2).$$

This visual discipline matters because clinical PK constantly moves between population, individual, and observation.

Symbol	Meaning
Y_{ij}	random observation, for example a future concentration.
y_{ij}	observed value after the assay reports a number.
$\boldsymbol{\psi}_i$	random individual parameter vector.
ψ_i	candidate or realized individual parameter value.
θ	population parameter when treated as an unknown fixed quantity.
$\boldsymbol{\theta}$	population parameter when treated as random in full Bayes.
$CL, V, AUC, AUMC, PK, PD$	fixed multi-letter PK/PD labels in equations.

2.2 Primary and secondary parameters

The course becomes clearer if we distinguish primary from secondary quantities. Primary parameters are closer to the system's mechanisms: CL, V, k_a, bioavailability F, distribution clearance Q, binding fractions, intrinsic clearance, organ blood flow. Secondary quantities are derived from them:

$$t_{1/2} = \frac{\log 2}{k_e}, \qquad k_e = \frac{CL}{V}, \qquad AUC = \frac{F\,Dose}{CL}.$$

Half-life is not a knob in the body. It is the visible consequence of clearance and volume. The same half-life can arise from high CL and high V, or from low CL and low V. This is why dose adjustment by half-life alone is often crude.

2.3 The four layers of a formula

Every formula has four layers:

(i) algebraic layer: the symbols balance;

(ii) unit layer: amount, volume, time, and concentration are consistent;

(iii) physiological layer: the signs and dependencies make biological sense;

(iv) decision layer: the equation answers the current clinical question.

For example, $AUC = Dose/CL$ is algebraically simple. But if the route is oral, the formula becomes $AUC = FDose/CL$. If CL is plasma clearance, organ extraction arguments require blood-to-plasma correction. If protein binding changes, total and unbound AUC may tell different clinical stories. A beautiful formula can still be the wrong formula for the decision.

3 PHC 607: The Concentration-Time Grammar

3.1 Amount, concentration, and compartments

The first conceptual split is amount versus concentration. Amount $A(t)$ is how much drug is in a compartment. Concentration $C(t)$ is amount divided by a volume-like scaling:

$$C(t) = \frac{A(t)}{V}.$$

The volume is not necessarily an anatomic bucket. It is a proportionality that turns amount into the measured concentration in the sampled matrix. A large apparent volume of distribution does not mean the body has a giant physical container. It means plasma concentration is small relative to total amount.

Compartment models are bookkeeping devices for mass balance. A compartment is a region that is assumed to be kinetically homogeneous for the question at hand. It can represent plasma, central extracellular fluid, tissue, effect site, gut, liver, target binding, or a mathematical delay. The model is judged by whether its compartments create the right predictions, not by whether every compartment has a literal wall.

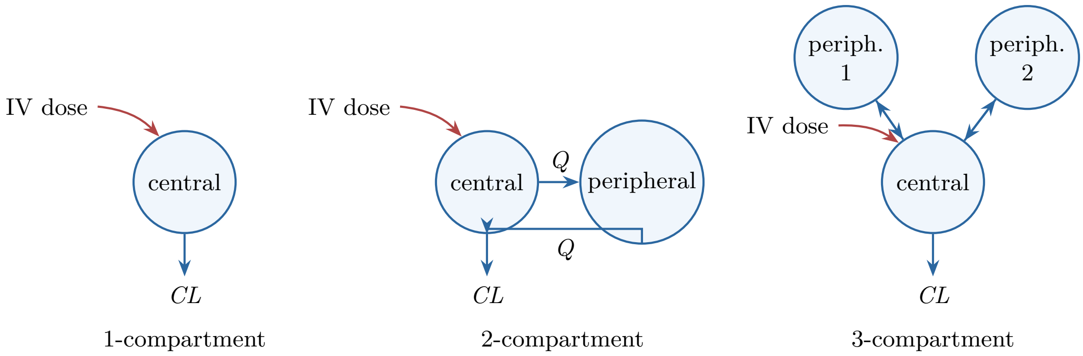


Figure 2: Compartmental models as mass-balance diagrams. The central compartment receives input and elimination; peripheral compartments add distribution time scales without adding systemic clearance unless explicitly modeled.

3.2 The first mass-balance equation

The one-compartment IV bolus model starts with

$$\frac{\mathrm{d}A(t)}{\mathrm{d}t} = -k_e A(t), \qquad A(0) = Dose.$$

The solution is

$$A(t) = Dose\, e^{-k_e t}, \qquad C(t) = C_0 e^{-k_e t}, \qquad C_0 = \frac{Dose}{V}.$$

This is the first clinical grammar: the intercept names distribution scale, the slope names elimination speed. But the intercept and slope are not independent mechanisms:

$$CL = k_e V.$$

The slope is not simply "renal function" or "metabolism." It is clearance divided by volume.

3.3 Zero-order and first-order as questions about rate

First-order elimination says the rate is proportional to amount:

$$\frac{\mathrm{d}A}{\mathrm{d}t} = -kA.$$

Zero-order elimination says the rate is approximately constant:

$$\frac{\mathrm{d}A}{\mathrm{d}t} = -R.$$

The clinical meaning is not "one is easy and one is hard." The clinical meaning is capacity. First-order kinetics is what a low-concentration approximation often looks like when elimination capacity is not stressed. Zero-order kinetics appears when the system is locally saturated or when input is controlled at a constant rate. PHC 609 will later reframe this using Michaelis–Menten elimination:

$$\frac{\mathrm{d}A}{\mathrm{d}t} = -\frac{V_{\max}C}{K_m + C}.$$

When $C \ll K_m$, the expression is approximately first-order; when $C \gg K_m$, it is approximately zero-order.

4 Clearance as the Main Invariant

4.1 Clearance is a proportionality, not a cleaned volume

Clearance is defined by

$$\text{rate of elimination} = CL\,C.$$

The phrase "volume cleared per time" is useful but dangerous. No physical volume of plasma is necessarily scrubbed clean. Clearance is a proportionality between concentration and elimination rate. It is the parameter that lets exposure connect to dose:

$$AUC_{0-\infty} = \frac{Dose}{CL} \quad \text{for IV bolus with linear elimination.}$$

If the drug is extravascular,

$$AUC_{0-\infty} = \frac{F\,Dose}{CL}.$$

Thus, for linear PK, exposure is mostly a clearance story. Volume shapes the time profile, but clearance controls total exposure.

4.2 Systemic, organ, blood, plasma, and unbound clearance

The first subtlety is reference concentration. A hepatic extraction model uses blood concentration entering and leaving the liver. Many clinical assays report plasma concentration. If C_b is blood concentration and C_p is plasma concentration, then

$$CL_b = CL_p \frac{C_p}{C_b}.$$

Confusing plasma clearance with blood clearance can make a drug seem to exceed organ blood flow, not because physiology is impossible, but because the denominator changed.

The second subtlety is binding. For many mechanistic arguments, unbound concentration is closer to the concentration available for transport, metabolism, receptor binding, and filtration. Total concentration is what we often measure, but unbound concentration is often what the enzyme or target sees:

$$C_u = f_u C.$$

The art is not to worship C_u blindly. The art is to ask which concentration the model equation is using.

4.3 Extraction ratio

Organ clearance can be written

$$CL_{\text{organ}} = Q\,E,$$

where Q is organ blood flow and E is extraction ratio. For liver,

$$E_H = \frac{C_{\text{in}} - C_{\text{out}}}{C_{\text{in}}}, \qquad CL_H = Q_H E_H.$$

High-extraction drugs are flow sensitive: changing Q_H changes clearance strongly. Low-extraction drugs are capacity and binding sensitive: changing f_u or intrinsic clearance matters more. This is not a memorized classification; it follows from the well-stirred model,

$$CL_H = \frac{Q_H f_u CL_{\text{int}}}{Q_H + f_u CL_{\text{int}}}.$$

If $f_u CL_{\text{int}} \gg Q_H$, then $CL_H \approx Q_H$. If $f_u CL_{\text{int}} \ll Q_H$, then $CL_H \approx f_u CL_{\text{int}}$.

5 Volume of Distribution and the Geography of Amount

5.1 Volume is a scaling

The apparent volume of distribution is

$$V = \frac{A}{C}.$$

The denominator is usually plasma or blood concentration. If a drug leaves plasma and partitions into tissues, plasma concentration can be small while total amount remains large, so V becomes large. This makes V a map of where amount hides relative to the sampled compartment.

The clinical use of V is immediate:

$$\textit{Loading Dose} = \frac{C_{\text{target}}\, V}{F}.$$

Loading dose is a distribution problem. Maintenance dose is a clearance problem. This single distinction prevents a lot of clinical confusion.

5.2 Protein binding does not have one universal consequence

Suppose plasma protein binding decreases, so f_u increases. What happens? It depends. The unbound concentration may initially rise, but clearance and distribution may also change. For a low-extraction hepatically cleared drug, total clearance may increase roughly with f_u. For a high-extraction drug, hepatic clearance may be flow limited and less sensitive to f_u. For a drug with a narrow therapeutic index where unbound concentration drives toxicity, a change in total concentration may be misleading.

The correct question is therefore not "does protein binding matter?" but

which concentration drives elimination, distribution, response, and measurement?

When these differ, total concentration becomes a shadow of several mechanisms.

5.3 Distribution rate versus distribution extent

Extent of distribution is about how much drug lives outside the sampled space at equilibrium. Rate of distribution is about how fast it gets there. A two-compartment model separates these:

$$\begin{aligned}
\frac{\mathrm{d}A_c}{\mathrm{d}t} &= -(CL + Q)\frac{A_c}{V_c} + Q\frac{A_p}{V_p}, \\
\frac{\mathrm{d}A_p}{\mathrm{d}t} &= Q\frac{A_c}{V_c} - Q\frac{A_p}{V_p}.
\end{aligned}$$

Here Q is distribution clearance. It controls exchange rate, not systemic elimination. A drug can have a large peripheral volume but slow equilibration, creating early high plasma concentrations followed by a long distribution tail.

6 Absorption and Bioavailability

6.1 Extent and rate are different questions

Absorption asks two questions:

how much gets in? how fast does it get in?

The first is extent, often summarized by bioavailability F. The second is rate, often summarized by k_a, $T_{\max}$, and $C_{\max}$. A formulation can preserve AUC while lowering $C_{\max}$. Another can increase $C_{\max}$ without changing total exposure. These are not cosmetic differences: toxicity, efficacy, and tolerability may depend on peak, trough, duration, or total exposure.

For first-order absorption into a one-compartment model,

$$C(t) = \frac{F\,Dose\,k_a}{V(k_a - k_e)}\left(e^{-k_e t} - e^{-k_a t}\right).$$

The curve rises because input initially exceeds elimination. It peaks when input and output balance. It falls when absorption has mostly finished and elimination dominates.

6.2 Bioavailability is a product of barriers

For oral dosing,

$$F = F_a F_G F_H,$$

where F_a is fraction absorbed across the gut wall, F_G is fraction escaping gut metabolism or efflux loss, and F_H is fraction escaping hepatic first pass. This factorization is a conceptual map, not always a directly identifiable statistical model. A low oral F could reflect solubility, permeability, degradation, transporter efflux, intestinal metabolism, hepatic extraction, or several together.

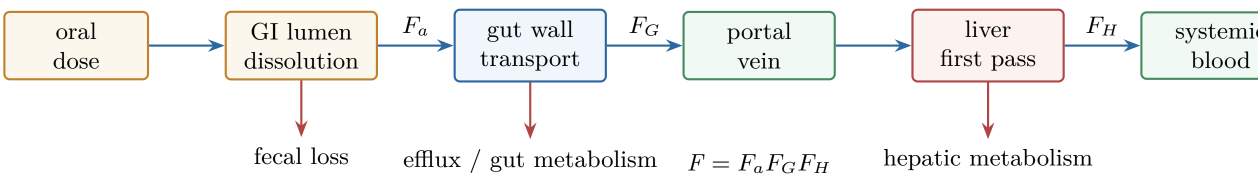


Figure 3: Oral bioavailability as a sequence of barriers. The figure abstracts the gut–portal–liver structure from the source notebooks into the factorization $F = F_a F_G F_H$.

6.3 Flip-flop kinetics

Normally $k_a > k_e$, so terminal decline reflects elimination. In flip-flop kinetics, $k_a < k_e$, so terminal decline reflects absorption. The terminal slope is then not the elimination slope. This is a beautiful cautionary example: the same visual straight line on a semi-log plot can mean different mechanisms depending on model structure.

The clinical consequence is serious. If the observed terminal half-life after an oral depot formulation is actually absorption-limited, using it to infer elimination half-life may overestimate how long drug remains once input stops.

7 Noncompartmental Analysis as Geometry

7.1 NCA describes the curve before explaining it

Noncompartmental analysis is often introduced as model-free. Better: NCA is low-model geometry plus a terminal exponential assumption. It reads the concentration-time curve by area, peak, time of peak, terminal slope, and moments:

$$AUC = \int_0^\infty C(t)\,\mathrm{d}t, \qquad AUMC = \int_0^\infty tC(t)\,\mathrm{d}t, \qquad MRT = \frac{AUMC}{AUC}.$$

The observed portion uses trapezoids. The unobserved tail uses λ_z:

$$AUC_{0-\infty} = AUC_{0-t_{\text{last}}} + \frac{C_{\text{last}}}{\lambda_z}.$$

Thus, NCA is not free of assumptions. It assumes the terminal portion chosen for λ_z is meaningful.

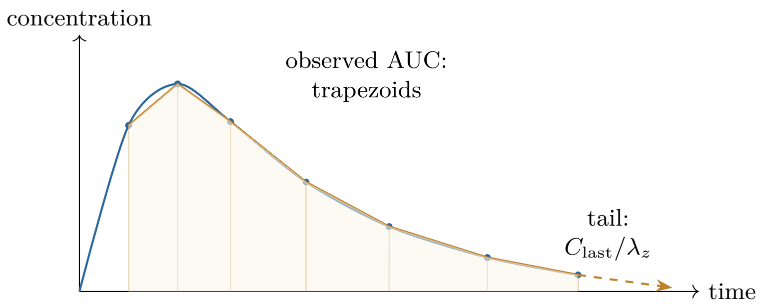


Figure 4: NCA as curve geometry. The observed area is numerical geometry; the terminal tail is a model assumption about λ_z.

7.2 AUC, Cmax, Tmax answer different questions

AUC is total exposure. $C_{\max}$ is peak intensity. $T_{\max}$ is a rough timing summary. For bioequivalence, AUC and $C_{\max}$ often carry the central evidence because they summarize extent and peak. For clinical toxicity, trough may matter more than peak. For concentration-dependent antibiotics, peak-to-MIC or AUC-to-MIC may be decisive. For time-dependent antibiotics, time above MIC matters.

The same curve contains many summaries. NCA is useful because it gives neutral summaries before committing to a structural explanation.

7.3 When NCA is enough and when it is not

NCA is often enough for single-dose exposure comparison, bioavailability, bioequivalence, and descriptive Phase 1 summaries. It is not enough when we need to predict unobserved regimens, covariate effects, sparse clinical sampling, nonlinear kinetics, delayed response, or patient-specific Bayesian updating. The transition from PHC 607 to PHC 608 is exactly this transition: from describing a curve to designing a regimen.

8 From Single Dose to Repeated Dosing

8.1 Superposition

For linear systems, multiple dosing can be built by superposition. If $C_1(t)$ is the concentration after one dose, then dosing every τ hours gives

$$C_n(t) = \sum_{m=0}^{n-1} C_1(t - m\tau) \quad \text{for terms with } t - m\tau \geq 0.$$

This is the mathematical reason why the first one-compartment curve is not a toy. Once linearity holds, it becomes a reusable impulse response.

8.2 Accumulation and steady state

For one-compartment IV bolus with first-order elimination, the accumulation factor is

$$R = \frac{1}{1 - e^{-k_e\tau}}.$$

Steady state is not a static concentration; it is a periodic pattern where each dosing interval repeats. Peaks, troughs, and average concentration stabilize:

$$C_{ss,\text{avg}} = \frac{F\,Dose}{CL\,\tau}.$$

Average steady-state concentration is a clearance result. Fluctuation is a half-life and dosing-interval result. This lets us separate two questions: how high is the average exposure, and how rough is the ride?

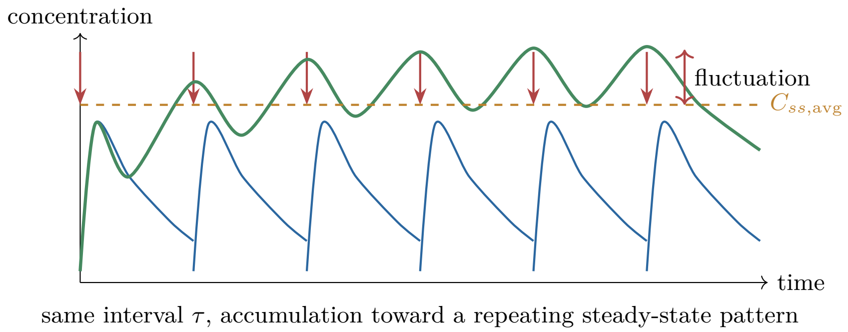


Figure 5: Multiple dosing by superposition. Repeated doses accumulate until the profile repeats; average exposure is clearance-controlled, while fluctuation is shaped by half-life and τ.

8.3 Time to steady state

Time to steady state is governed by k_e, not by dose. It takes about four to five half-lives to get close to steady state for linear first-order kinetics. Raising the maintenance dose raises the plateau; it does not make the system approach the plateau faster. A loading dose can jump the system near the desired amount because loading is a volume problem:

$$LD = \frac{C_{\text{target}}\, V}{F}.$$

Maintenance is then a clearance problem:

$$MD/\tau = \frac{C_{\text{target}}\, CL}{F}.$$

9 Infusion as Controlled Input

9.1 Zero-order input with first-order elimination

For constant infusion rate R_0,

$$\frac{\mathrm{d}A}{\mathrm{d}t} = R_0 - k_e A.$$

The solution during infusion is

$$C(t) = \frac{R_0}{CL}\left(1 - e^{-k_e t}\right).$$

The steady-state concentration is

$$C_{ss} = \frac{R_0}{CL}.$$

This is one of the cleanest clinical equations in PK. If target concentration is known and clearance is known, rate follows directly:

$$R_0 = C_{ss} CL.$$

9.2 Stopping the infusion

If infusion stops at time T, concentration declines from whatever level was reached:

$$C(t) = C(T)e^{-k_e(t-T)}, \qquad t \geq T.$$

This simple piecewise thinking is a major clinical skill. During infusion, input and elimination both act. After infusion, elimination acts alone. The equation you use depends on where the patient is in the regimen.

9.3 Short infusions

Many clinical regimens are neither bolus nor continuous infusion. A short infusion is a controlled input over a finite duration, repeated every τ. It combines the infusion rise and post-infusion decline. The important habit is to label the clock: time since start of infusion, time since end of infusion, and time within the dosing interval are not the same.

10 Two-Compartment Thinking

10.1 Why biexponential curves happen

After IV bolus, many drugs show a fast early decline and slower later decline:

$$C(t) = Ae^{-\alpha t} + Be^{-\beta t}.$$

The fast phase is not necessarily rapid elimination. It may be distribution out of central compartment. The slow phase is not necessarily a separate depot. It is the terminal eigenmode of the coupled system.

This is where linear algebra quietly enters clinical PK. The exponents α, β are macro-rate constants arising from the eigenvalues of the compartment system. The micro-rate constants k_{12}, k_{21}, k_{10} describe exchange and elimination. The observed curve is a mixture of modes.

10.2 Volumes in a two-compartment world

Several volumes appear:

$$V_c, \qquad V_{ss}, \qquad V_\beta.$$

V_c is the central volume relevant to initial concentration after IV bolus. V_{ss} is the steady-state extent of distribution. V_β connects clearance to terminal slope:

$$V_\beta = \frac{CL}{\beta}.$$

These volumes answer different questions. Using the wrong one can distort loading dose, half-life interpretation, or tissue accumulation intuition.

10.3 Central versus peripheral site of action

If the site of action is central, early concentration matters. If the site of action is peripheral, effect may lag behind plasma concentration because peripheral equilibration is slow. This is one reason PK/PD cannot stop at plasma curves. A clinically relevant response may be driven by an effect-site concentration, receptor occupancy, cell turnover, or downstream biomarker.

11 Metabolites and Sequential Systems

11.1 Parent to metabolite

For a parent drug A_p forming a metabolite A_m,

$$\frac{\mathrm{d}A_p}{\mathrm{d}t} = -k_p A_p, \qquad \frac{\mathrm{d}A_m}{\mathrm{d}t} = f_m k_p A_p - k_m A_m.$$

The metabolite curve rises while formation exceeds elimination and falls when formation can no longer sustain it. If $k_m \gg k_p$, metabolite kinetics are formation-rate limited. If $k_p \gg k_m$, they are metabolite-elimination limited.

11.2 Active metabolites

An active metabolite changes the dose-response story. The parent concentration may fall while pharmacological activity persists. A response model may need parent, metabolite, or both:

$$E(t) = E_0 + \frac{E_{\max} C_p(t)}{EC_{50} + C_p(t)} + \frac{E^m_{\max} C_m(t)}{EC^m_{50} + C_m(t)}.$$

This is a simple formula, but it carries a clinical warning: measuring only parent may miss part of pharmacology.

12 Nonlinear Pharmacokinetics

12.1 Symptoms of nonlinearity

Nonlinear PK appears when dose changes do not produce proportional exposure:

$$\frac{AUC_2}{AUC_1} \neq \frac{Dose_2}{Dose_1}.$$

Other symptoms include dose-dependent half-life, clearance changing with concentration, time-dependent kinetics, unexpected accumulation, and terminal slopes that change across doses.

Nonlinearity is not a moral failure of a model. It is often a biological clue: saturable absorption, solubility limit, saturable first-pass metabolism, capacity-limited elimination, saturable plasma or tissue binding, transporter saturation, target-mediated disposition, autoinduction, autoinhibition, or disease-modulated physiology.

12.2 Michaelis–Menten elimination

For capacity-limited elimination,

$$\frac{\mathrm{d}A}{\mathrm{d}t} = -\frac{V_{\max}C}{K_m + C}.$$

The apparent clearance is

$$CL_{\mathrm{app}}(C) = \frac{V_{\max}}{K_m + C}.$$

As concentration increases, apparent clearance decreases. This is why a small dose increase can produce a large concentration increase near saturation.

12.3 Phenytoin as a clinical warning

Phenytoin is a standard teaching example because it makes the clinical danger visible. At steady state,

$$R_{\mathrm{in}} = \frac{V_{\max}C_{ss}}{K_m + C_{ss}}.$$

Solving for C_{ss},

$$C_{ss} = \frac{K_m R_{\mathrm{in}}}{V_{\max} - R_{\mathrm{in}}}.$$

If R_{in} approaches $V_{\max}$, concentration climbs sharply. The denominator is the danger. Linear dose adjustment fails because there is no constant clearance.

13 Therapeutic Drug Monitoring

13.1 TDM as Bayesian updating before software

Therapeutic drug monitoring is often taught as plug-in equations. But the deeper logic is Bayesian:

$$\text{prior patient expectation} + \text{measured concentration} \longrightarrow \text{updated individual parameters} \longrightarrow \text{next dose.}$$

Even a simple peak/trough aminoglycoside calculation implicitly combines prior model structure with observed data. Modern Bayesian TDM simply makes the hierarchy explicit.

13.2 Peak, trough, AUC, and target

Different drugs ask for different monitoring targets. Aminoglycosides may emphasize peak-to-MIC and trough safety. Vancomycin increasingly emphasizes AUC-guided dosing. Antiepileptics may use trough ranges, but interpretation depends on adherence, timing, protein binding, interacting drugs, and symptoms. Immunosuppressants require exposure control under substantial variability.

The correct TDM question is not "is the level in range?" It is:

$$\text{given this level, timing, dose history, patient physiology, and target, what should happen next?}$$

13.3 Measurement is part of the model

Assay error, sampling time error, missed doses, infusion timing, and documentation errors can dominate the apparent PK story. A concentration drawn during distribution may not represent trough. A sample labeled as trough may have been drawn early. A patient may have missed a dose. The clinical model includes the hospital workflow.

14 Special Populations

14.1 Pediatrics

Pediatric dosing is not adult dosing divided by body weight. Maturation changes renal function, hepatic enzymes, plasma proteins, body water, fat fraction, gastric pH, gastric emptying, and blood flow. Size matters, but maturation matters too. A common population model separates them:

$$CL_i = CL_{\text{adult}} \left(\frac{WT_i}{70} \right)^{0.75} Mat(PMA_i) e^{\eta_{CL,i}}.$$

The allometric exponent captures size scaling; the maturation function captures developmental physiology.

14.2 Elderly patients

Elderly patients challenge single-parameter thinking. Renal function may decline, hepatic blood flow may change, body composition shifts, albumin may be lower, comedications increase, homeostatic reserve declines, and pharmacodynamic sensitivity may increase. A normal concentration can still be clinically excessive if response sensitivity changed.

14.3 Obesity

Obesity forces the distinction between loading and maintenance. Loading dose depends on distribution; maintenance dose depends on clearance. Total body weight, ideal body weight, adjusted body weight, lean body weight, and body surface area are not interchangeable. The right body-size descriptor depends on whether the drug distributes into adipose, lean tissue, extracellular water, or a mixture.

14.4 Renal and hepatic impairment

Renal impairment affects filtration, secretion, reabsorption, protein binding, metabolites, and sometimes nonrenal clearance through uremic effects on enzymes and transporters. Hepatic impairment affects blood flow, intrinsic clearance, protein binding, biliary excretion, shunting, and first-pass extraction. Dose adjustment cannot be reduced to one lab value, although creatinine clearance, eGFR, Child–Pugh, bilirubin, albumin, INR, and disease context are useful summaries.

15 Drug-Drug Interactions

15.1 Enzymes, transporters, and exposure

A drug-drug interaction is a change in the system map. Inhibition decreases a pathway's capacity. Induction increases expression or activity over time. Transporter inhibition can alter absorption, hepatic uptake, renal secretion, brain entry, or biliary excretion. The exposure consequence depends on fraction affected:

$$\frac{AUC_{\text{with}}}{AUC_{\text{without}}} \approx \frac{1}{(1 - f_m) + f_m/R},$$

where f_m is fraction of clearance through the pathway and R is the residual activity ratio. If a pathway contributes little, even strong inhibition may have modest effect. If a pathway dominates, modest inhibition may matter.

15.2 Time course of interaction

Not all interactions appear immediately. Competitive inhibition can be rapid. Mechanism-based inhibition may develop as enzyme is inactivated. Induction requires synthesis and degradation dynamics. Thus, the time course of interaction can be a PK/PD model of enzyme activity:

$$\frac{\mathrm{d}E_{\text{enz}}}{\mathrm{d}t} = k_{\text{syn}}(1 + \mathit{Ind}(C)) - k_{\text{deg}}E_{\text{enz}}.$$

The observed drug concentration is downstream of this enzyme state.

16 Biologics: A Different PK Intuition

16.1 Small molecules and biologics differ in scale

Biologics are not merely large small molecules. Size, structure, route, target binding, proteolysis, FcRn recycling, immunogenicity, and tissue access create different PK intuitions. Oral absorption is usually not the default. Distribution may be limited by convection, neonatal Fc receptor recycling, binding, and tissue penetration. Clearance may involve proteolysis, target-mediated elimination, immune complexes, or cellular processing rather than CYP metabolism.

16.2 Antibodies

Monoclonal antibodies often have small central volumes, slow distribution, long half-lives, and nonlinear clearance at low concentrations when target-mediated elimination is important. FcRn recycling protects IgG from lysosomal degradation and contributes to long persistence. But target binding can create nonlinear elimination:

$$D + R \underset{k_{\text{off}}}{\overset{k_{\text{on}}}{\rightleftharpoons}} DR \xrightarrow{k_{\text{int}}} \varnothing.$$

At low doses, target-mediated clearance can dominate. At high doses, target pathways saturate and nonspecific clearance may dominate.

16.3 ADCs

Antibody-drug conjugates are systems, not single molecules. The measured analyte might be conjugated antibody, total antibody, free payload, released payload metabolite, or antigen-bound complex. Linker stability, drug-antibody ratio, antigen expression, internalization, payload permeability, bystander killing, and off-target toxicity all matter. A single concentration curve can hide multiple chemical and biological entities.

16.4 Oligonucleotides and cell therapies

ASOs, siRNAs, and cell therapies further stretch standard PK. Oligonucleotides involve tissue uptake, endosomal escape, chemical modification, protein binding, renal handling, and long tissue residence. CAR-T therapies expand, contract, persist, traffic, and respond to antigen. Their “PK” may be cell count or vector copies, and their “dose” may start a living dynamical system.

17 PHC 609: The Advanced Modeling Turn

17.1 Model of system versus model of data

An advanced PK/PD course becomes easier when we separate model of system from model of data. A system model says how amounts, concentrations, receptors, biomarkers, cells, or disease states evolve. A data model says how observations are generated from those latent states:

$$\frac{\mathrm{d}x}{\mathrm{d}t} = f(x, u, \psi, t),$$
$$y_{ij} = g(x(t_{ij}), \psi) + \epsilon_{ij}.$$

The first line is biology and pharmacology. The second is measurement. Confusing them leads to poor model criticism.

17.2 Laplace transforms and SHAM

For linear time-invariant PK systems, Laplace transforms reveal how input becomes output:

$$X(s) = G(s)U(s).$$

Poles determine slopes. Gains determine heights and areas. The SHAM mnemonic – slope, height, area, moment – is a way to inspect curves without forgetting the system behind them. A concentration profile is not only a graph; it is a compressed signature of rate constants, volumes, clearance, input, and observation.

17.3 Numerical solution is not a fallback

Closed forms are pedagogically beautiful, but numerical ODE solution is not second-class. PBPK, TMDD, QSP, nonlinear PD, disease progression, and complex dosing histories usually require numerical integration. The key is not whether the solution is analytical; the key is whether the model is verified, identifiable enough for the question, and numerically stable under the scenarios used for decision.

18 Fitting PK Models

18.1 Objective functions

Fitting means choosing parameters whose predictions are close to data under a stated error model. Ordinary least squares assumes constant additive variance:

$$\min_{\psi} \sum_{j} (y_j - \hat{y}_j(\psi))^2.$$

Weighted least squares accounts for changing variance:

$$\min_{\psi} \sum_{j} w_j (y_j - \hat{y}_j(\psi))^2.$$

A likelihood-based approach writes the observation law explicitly:

$$y_j \mid \psi \sim \mathcal{N}(\hat{y}_j(\psi), \sigma_j^2).$$

The likelihood is not a technicality. It is the model of how assays, residual variability, and misspecification create discrepancies.

18.2 Residual error models

Common residual models include additive,

$$y = \hat{y} + \epsilon,$$

proportional,

$$y = \hat{y}(1 + \epsilon),$$

and combined forms. On the log scale, multiplicative error becomes additive:

$$\log y = \log \hat{y} + \epsilon.$$

Choosing the residual model changes which deviations are treated as surprising. This is why residual plots are not decorative; they are direct checks of the observation model.

18.3 Local minima and parameter correlation

PK model fitting often has correlated parameters. CL and V can compensate through half-life; k_a and k_e can trade off when absorption is poorly sampled; $V_{\max}$ and K_m can be weakly identified outside the concentration range that tests saturation. A best fit can be numerically precise but scientifically fragile. Standard errors, profiles, bootstrap, posterior samples, and simulation checks all ask whether the fitted parameter is actually learned.

19 Population Pharmacokinetics

19.1 The hierarchy

Population PK begins when individual parameters vary:

$$\psi_i = h(c_i, \theta, \eta_i), \qquad \eta_i \sim \mathcal{N}(0, \Omega).$$

A typical lognormal clearance model is

$$CL_i = \theta_{CL} \left(\frac{WT_i}{70} \right)^{\theta_{\mathrm{WT}}} e^{\eta_{CL,i}}.$$

The observation model might be

$$y_{ij} = \hat{y}(t_{ij}; \psi_i, u_i)(1 + \epsilon_{ij}).$$

This is the same hierarchy from the earlier Bayesian PopPK note, now placed inside the clinical PK/PD course.

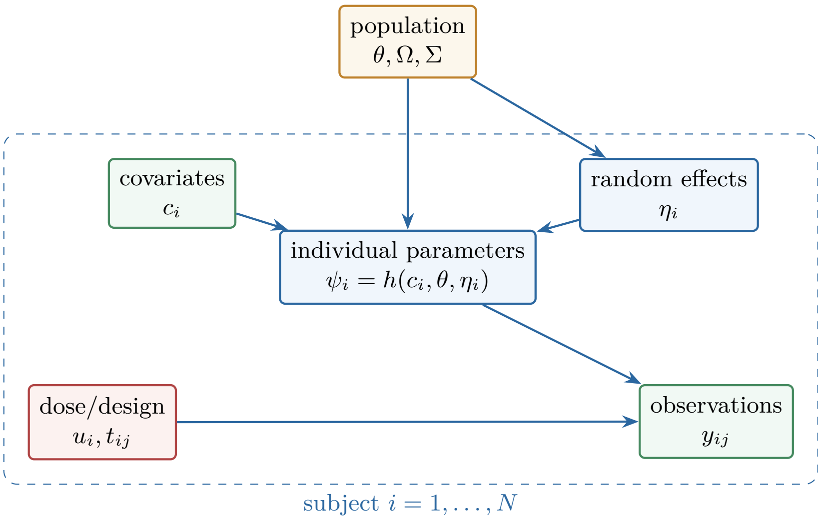


Figure 6: Population PK hierarchy. NONMEM, SAEM, and Bayesian TDM are different computational cultures around this same latent-individual-parameter graph.

19.2 Fixed effects, random effects, residual effects

Fixed effects describe typical values and covariate relationships. Random effects describe unexplained interindividual variability. Residual effects describe observation-level mismatch. The three are not interchangeable. If a covariate is omitted, random variability may inflate. If a structural model is wrong, residual error may inflate. If data are sparse, individual random effects may shrink toward zero. Diagnostics must ask which layer is absorbing the mismatch.

19.3 NONMEM as an inference dialect

NONMEM's THETA, ETA, and EPS syntax is not a separate theory. It is a dialect for the hierarchy:

$$\begin{array}{lcl} \text{THETA} & \leftrightarrow & \text{population typical values and covariate effects,} \\ \text{ETA} & \leftrightarrow & \text{individual deviations,} \\ \text{EPS} & \leftrightarrow & \text{residual observation noise,} \\ \Omega, \Sigma & \leftrightarrow & \text{variance components.} \end{array}$$

The objective function approximates a marginal likelihood where individual random effects have been integrated out. FO, FOCE, Laplacian, SAEM, and MCMC are different computational routes through the same conceptual problem.

20 Bayesian Dose Individualization

20.1 Individual posterior

Given a population model and a patient's observed concentrations,

$$p(\psi_i \mid y_i, c_i, u_i, \theta) \propto p(y_i \mid \psi_i, u_i)p(\psi_i \mid c_i, \theta).$$

This is the mathematical heart of Bayesian TDM. The prior-like term is the population model. The likelihood is the patient's measured data. The posterior is the updated individual PK state.

20.2 Decision rule

The dose is not chosen from a point estimate alone. In principle, the next regimen a should minimize expected loss:

$$a^* = \arg\min_a \mathbb{E}\left[\mathcal{L}(a, \tilde{y}, \tilde{r}) \mid y_i, c_i, u_i\right].$$

Here $\tilde{y}$ and $\tilde{r}$ are future exposure and response. A trough target, AUC target, toxicity constraint, or probability-of-target-attainment rule is a practical version of this expected-loss calculation.

20.3 Why posterior uncertainty matters

Two patients may have the same posterior mean CL but different uncertainty. A robust dosing decision may differ if one posterior is tight and the other is broad. Sparse sampling, uncertain adherence, renal instability, rapidly changing disease, and nonlinear kinetics widen the posterior predictive distribution. A good Bayesian report should show not only the recommended dose but also the uncertainty in achieving the target.

21 PBPK: Physiology as Model Architecture

21.1 Whole-body structure

PBPK models replace abstract compartments with physiological tissues connected by blood flow:

$$\frac{\mathrm{d}A_j}{\mathrm{d}t} = Q_j C_{\mathrm{art}} - Q_j \frac{C_j}{K_{p,j}} - Elim_j.$$

The parameters now include tissue volumes, blood flows, partition coefficients, permeability, enzyme abundances, transporter activities, plasma binding, and route-specific absorption physiology.

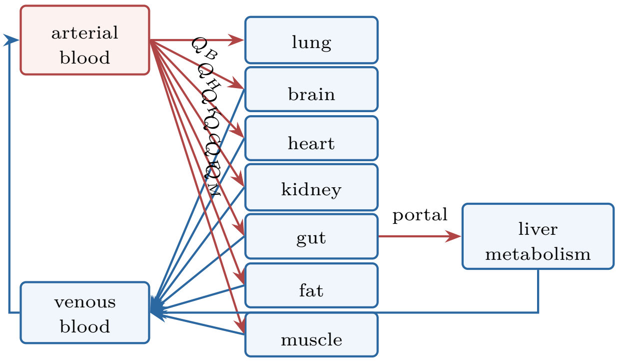


Figure 7: A stripped-down PBPK circulation. Whole-body PBPK replaces abstract compartments with blood-flow-connected organs, while still preserving mass balance and observation maps.

21.2 What PBPK is good for

PBPK is especially useful when extrapolation depends on physiology: first-in-human translation, pediatric scaling, pregnancy, renal/hepatic impairment, drug-drug interactions, tissue exposure, transporter hypotheses, and virtual populations. Its power is not that it has many compartments. Its power is that those compartments are anchored to measurable physiological quantities.

21.3 What PBPK is not

PBPK is not automatically more true because it is more detailed. A complex model can hide uncertainty behind many parameters. The key questions are:

which prediction is needed, which parameters control it, and what evidence supports those parameters?

This is the same discipline as simpler PK, only with more moving parts.

22 Target-Mediated Drug Disposition

22.1 The basic mechanism

TMDD occurs when binding to a pharmacological target changes disposition. A minimal model is

$$\begin{aligned}\frac{\mathrm{d}D}{\mathrm{d}t} &= -k_{\mathrm{on}} DR + k_{\mathrm{off}} DR_c - CL_{\mathrm{lin}} D,\\ \frac{\mathrm{d}R}{\mathrm{d}t} &= k_{\mathrm{syn}} - k_{\mathrm{deg}} R - k_{\mathrm{on}} DR + k_{\mathrm{off}} DR_c,\\ \frac{\mathrm{d}DR_c}{\mathrm{d}t} &= k_{\mathrm{on}} DR - (k_{\mathrm{off}} + k_{\mathrm{int}}) DR_c.\end{aligned}$$

This model links PK and pharmacology because the target is both a sink and a mediator of effect.

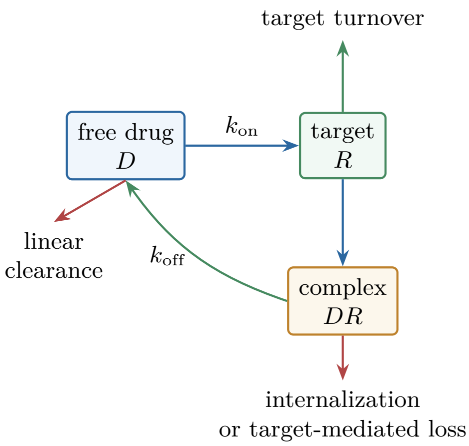


Figure 8: Target-mediated drug disposition. TMDD makes the pharmacological target part of the disposition model, so PK and PD become coupled rather than sequential.

22.2 Small molecule versus large molecule TMDD

For biologics, TMDD often appears because target binding and internalization are important relative to nonspecific clearance. For small molecules, target binding may be rapid, tissue-localized, or not a major elimination route, so TMDD may be harder to see or mechanistically different. The visual symptom – nonlinear exposure – is not enough. We need mechanism, target biology, concentration range, and perturbation evidence.

22.3 Reduced TMDD models

Full TMDD models can be difficult to identify. Quasi-equilibrium and quasi-steady-state approximations reduce the system when binding dynamics are fast relative to disposition. These approximations are not shortcuts by laziness; they are scientific statements about time scales. If the time-scale statement is wrong, the reduced model can mislead.

23 Pharmacodynamics

23.1 Direct effect

The simplest PD model maps concentration to effect immediately:

$$E(C) = E_0 + \frac{E_{\max} C}{EC_{50} + C}.$$

The direct-effect model is appropriate when measured concentration equilibrates rapidly with the biophase and response is immediate. It teaches three ideas: baseline, maximum effect, and potency. But many clinical responses are not immediate.

23.2 Effect compartment

When plasma concentration and effect show hysteresis, an effect compartment can introduce equilibration delay:

$$\frac{\mathrm{d}C_e}{\mathrm{d}t} = k_{e0}(C_p - C_e), \qquad E(t) = E(C_e(t)).$$

The effect compartment is not necessarily a physical compartment. It is a kinetic representation of delay between measured concentration and biophase concentration.

23.3 Indirect response

If drug changes production or loss of a response variable, an indirect response model is more natural:

$$\frac{\mathrm{d}R}{\mathrm{d}t} = k_{\text{in}}\, S(C) - k_{\text{out}} R$$

or

$$\frac{\mathrm{d}R}{\mathrm{d}t} = k_{\text{in}} - k_{\text{out}}\, I(C) R.$$

This class explains delayed onset, rebound, tolerance-like behavior, and biomarker turnover. The response time scale is now governed by k_{out}, not only by drug half-life.

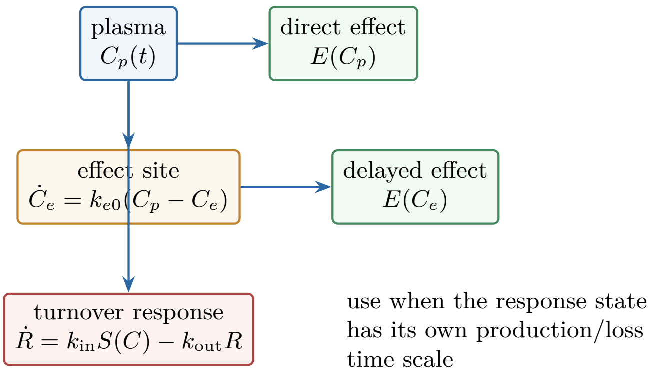


Figure 9: Three PD delay grammars. Direct effect, effect compartment, and turnover models answer different reasons why response and plasma concentration may disconnect.

24 Exposure-Response and Clinical Development

24.1 Learning versus confirming

Clinical development alternates between learning and confirming. Early studies learn tolerability, PK, dose range, and target engagement. Middle studies learn exposure-response and select dose. Pivotal studies confirm benefit-risk. Exposure-response analysis prevents dose selection from becoming a ritual of pairwise dose comparisons.

24.2 Dose selection

The dose should be justified by a chain:

$$\text{dose} \rightarrow \text{exposure} \rightarrow \text{target engagement} \rightarrow \text{biomarker} \rightarrow \text{clinical response} \rightarrow \text{safety.}$$

Not every program has every link. But the missing links should be named. Model-informed drug development is disciplined storytelling with equations, not curve fitting for decoration.

24.3 Project Optimus and oncology

Oncology historically tolerated maximum tolerated dose logic, especially for cytotoxic agents. Targeted therapies and immunotherapies often require more nuanced dose optimization. If efficacy saturates before toxicity, the highest tolerated dose may not be the best dose. PK/PD and exposure-response become central to choosing a dose that balances activity, tolerability, adherence, and long-term benefit.

25 QSP and Systems Pharmacology

25.1 When PK/PD becomes a system

QSP appears when the response cannot be represented by one turnover variable or one effect-site delay. Immune cells, cytokines, tumor burden, receptor trafficking, signaling networks, feedback loops, and disease progression may all interact. The state vector becomes

$$\frac{\mathrm{d}x}{\mathrm{d}t} = f(x, u, \theta, t), \qquad y = g(x, \theta, t) + \epsilon.$$

PK is one part of the state system, not the whole model.

25.2 Platform models

Platform QSP models try to reuse a mechanistic scaffold across related drugs or targets: bispecific antibodies, ADCs, siRNAs, immuno-oncology combinations. The promise is learning across programs. The danger is carrying forward assumptions beyond their evidence. Reuse is powerful only when assumptions travel with uncertainty.

25.3 Credibility

A QSP model is credible relative to a question of interest. It is not globally credible. For dose selection, the model must be credible for exposure-response in the relevant population and dose range. For mechanism exploration, qualitative dynamics may suffice. For regulatory support, documentation, verification, validation, sensitivity analysis, and uncertainty quantification become part of the model itself.

26 A Three-Course Learning Route

26.1 PHC 607: name the objects

Start by learning to read a concentration-time curve:

$$C_{\max}, \quad T_{\max}, \quad AUC, \quad \lambda_z, \quad t_{1/2}, \quad CL, \quad V, \quad F.$$

Then attach those summaries to LADME: liberation, absorption, distribution, metabolism, excretion. Do not rush to software. Learn what a slope means, what an intercept means, what area means, what a moment means, and what units reveal.

26.2 PHC 608: design the regimen

Next, learn to use the objects:

$$LD = \frac{C_{\text{target}}\,V}{F}, \qquad MD/\tau = \frac{C_{\text{target}}\,CL}{F}.$$

Study multiple dosing, infusion, oral absorption, two-compartment kinetics, nonlinear elimination, special populations, TDM, and drug interactions. The guiding question is always: what would I change in the next regimen, and why?

26.3 PHC 609: build and criticize models

Finally, learn model-based pharmacology:

$$\text{ODE} \to \text{likelihood} \to \text{population hierarchy} \to \text{prediction} \to \text{model criticism}.$$

This includes numerical integration, fitting, nonlinear PK, PBPK, PopPK, NONMEM, Bayesian updating, TMDD, direct and indirect PD, exposure-response, and QSP. The goal is not complexity. The goal is choosing the simplest model that can support the decision without hiding the important uncertainty.

27 Worked Mini-Cases

27.1 Case 1: loading and maintenance

A patient needs rapid achievement of a target concentration C^*. The model suggests $V = 40\ L$, $CL = 5\ L/h$, and $F = 1$. The loading dose is

$$LD = C^* V.$$

The maintenance infusion rate is

$$R_0 = C^* CL.$$

If V doubles but CL is unchanged, the loading dose doubles while maintenance rate remains the same. If CL doubles but V is unchanged, maintenance rate doubles while loading dose remains the same. This is the cleanest way to separate distribution from elimination.

27.2 Case 2: oral exposure falls

Suppose oral AUC falls after coadministration with another drug. The equation

$$AUC = \frac{F_a F_G F_H\ Dose}{CL}$$

creates hypotheses: reduced dissolution, reduced permeability, increased efflux, induced gut metabolism, induced hepatic metabolism, increased systemic clearance, or adherence/timing error. The curve shape helps prioritize. Lower $C_{\max}$ with similar terminal slope suggests absorption or bioavailability. Lower exposure with changed terminal slope may suggest systemic clearance.

27.3 Case 3: sparse TDM

A patient has one concentration after several doses. Alone, that concentration cannot identify CL, V, k_a, adherence, and residual error. A Bayesian population model makes the problem solvable by borrowing information:

$$p(\psi_i \mid y_i) \propto p(y_i \mid \psi_i) p(\psi_i).$$

The single concentration updates a prior distribution rather than creating physiology from nothing. This is why the population model is not optional decoration in sparse clinical dosing.

27.4 Case 4: biologic nonlinear clearance

A monoclonal antibody shows more-than-dose-proportional exposure as dose increases. One explanation is target-mediated clearance saturation. At low dose, target binding and internalization remove drug efficiently. At high dose, target pathways saturate, and linear nonspecific clearance dominates. The next experiment should not only fit a curve; it should ask whether target expression, receptor occupancy, knock-out data, or displacement evidence supports the mechanism.

28 Reading the Buffalo Notebooks Productively

28.1 What to extract

When reading the Buffalo materials, do not copy the order mechanically. Extract four things from each topic:

(i) the clinical question,

(ii) the state variables and parameters,

(iii) the forward prediction equation,

(iv) the decision or diagnostic enabled by the model.

For absorption, the clinical question is exposure after extravascular dosing. For two-compartment models, it is early versus late disposition and tissue equilibration. For nonlinear PK, it is dose adjustment under capacity. For TMDD, it is target binding as both mechanism and disposition. For QSP, it is system-level prediction under feedback.

28.2 What to avoid

Avoid treating headings as isolated facts. Absorption, distribution, elimination, and response are not chapters in nature. They are interfaces in the same dynamical system. A transporter can affect absorption, hepatic uptake, renal secretion, and tissue distribution. Protein binding can affect distribution, clearance, and measurement. A biomarker delay can masquerade as slow distribution. Clinical pharmacology is hard because the same observed curve can have several mechanistic explanations.

28.3 The mature reading loop

For each chapter, run this loop:

story $\rightarrow$ mass balance $\rightarrow$ parameter meaning $\rightarrow$ data model $\rightarrow$ diagnostic $\rightarrow$ decision.

If a topic cannot be placed on this loop, it is not yet understood.

29 Expanded Atlas I: Reading a PK Curve

29.1 The first pass: draw before calculating

The first skill in clinical pharmacokinetics is visual. Before calculating AUC, CL, or $t_{1/2}$, draw the curve and label five regions: input, peak, distribution, terminal decline, and unobserved tail. The curve is a narrative. A sharp rise says input is fast relative to elimination. A slow rise says absorption or infusion dominates. A sharp early drop after IV bolus may be distribution. A straight terminal semi-log line says one exponential mode is dominating the measured window.

The most useful first-pass questions are:

(i) Where is input still happening?

(ii) Which part is safe to treat as terminal?

(iii) Does the route make the terminal slope interpretable as elimination?

(iv) Are the sampling times dense enough around $C_{\max}$?

(v) Is the tail large enough that extrapolation could dominate AUC?

These questions prevent numerical summaries from pretending to be more certain than the design permits.

29.2 Slope, height, area, moment

The SHAM habit is a useful bridge between NCA and model-based thinking:

slope height area moment.

Slope asks about rates. Height asks about scale. Area asks about total exposure. Moment asks about residence time and timing. For an IV bolus one-compartment model:

$$C(t) = \frac{Dose}{V} e^{-(CL/V)t}.$$

The height at time zero is controlled by dose and volume. The slope is controlled by CL/V. The area is controlled by dose and clearance. The moment combines exposure and time, giving $MRT = V/CL$ in the simplest case.

29.3 A minimal curve reading protocol

For any profile, run this protocol:

(i) Identify route and input duration.

(ii) Confirm dose, units, matrix, and sampling clock.

(iii) Mark $C_{\max}, T_{\max}, C_{\text{last}}, t_{\text{last}}$.

(iv) Inspect terminal phase on a semi-log plot.

(v) Estimate λ_z only after visual inspection.

(vi) Compute AUC and extrapolated fraction.

(vii) Ask which clinical target uses $AUC, C_{\max}, C_{\text{trough}}$, time above threshold, or response.

This protocol is intentionally slow. Speed without this map is how one obtains precise but wrong PK.

30 Expanded Atlas II: Units, Signs, and Sanity Checks

30.1 Units are part of the equation

Every PK equation should survive unit inspection. If A is amount and C is concentration,

$$C = \frac{A}{V}$$

requires V to have volume units. If CL has volume per time and C has amount per volume, then CLC has amount per time, which can appear as an elimination rate. This is not bookkeeping trivia. Many modeling mistakes are unit mistakes disguised as biology.

For infusion,

$$\frac{\mathrm{d}A}{\mathrm{d}t} = R_0 - CLC = R_0 - \frac{CL}{V}A.$$

Both terms must be amount per time. If R_0 is entered in *mg/h* but CL is in *L/day*, the model may run and still be nonsense.

30.2 Signs reveal mass balance

Internal transfers cancel globally. In a two-compartment model, movement from central to peripheral is a loss for central and a gain for peripheral:

$$-QC_c \quad \text{in central,} \qquad +QC_c \quad \text{in peripheral.}$$

The same term should not have the same sign in both compartments unless mass is being created or destroyed. Elimination terms should be negative from the drug amount equations. Input terms should be positive unless modeling removal or dialysis.

30.3 Numerical sanity before inference

Before fitting, simulate easy cases:

- zero dose should give zero drug concentration unless baseline drug exists;
- zero clearance should prevent elimination;
- larger dose should raise concentration under linear PK;
- larger clearance should lower AUC;
- larger volume should lower initial IV bolus concentration and lengthen half-life if clearance is fixed.

These checks are simple, but they catch sign errors, unit errors, wrong compartment dosing, wrong output scaling, and bad initial conditions.

31 Expanded Atlas III: Absorption as a Sequence of Losses

31.1 The oral dose is a chain

An oral dose must dissolve, survive the lumen, cross the gut wall, escape efflux and gut metabolism, pass through portal circulation, survive hepatic extraction, and then enter systemic blood. The factorization

$$F = F_a F_G F_H$$

is a thinking tool. It says oral exposure is not one event. It is a sequence of barriers.

A low F therefore has many possible explanations. A formulation problem might reduce F_a. A food effect might delay gastric emptying or change solubility. A transporter inducer might reduce net absorption. A CYP3A inducer might reduce F_G or increase systemic clearance. A high-extraction liver drug might have low F_H. One observed AUC ratio rarely identifies which link failed without additional evidence.

31.2 Food, milk, tea, and transporters

Classic teaching examples are valuable because they show mechanisms rather than names. Milk can reduce absorption of certain drugs by chelation. Food can delay or enhance absorption depending on dissolution, gastric emptying, bile secretion, and transporter effects. Green tea reducing nadolol exposure points toward transporter-mediated absorption. The general lesson is:

extravascular PK is formulation + physiology + transport + first pass + systemic clearance.

If the terminal slope is unchanged but $C_{\max}$ and AUC fall, absorption or bioavailability is suspicious. If terminal slope changes too, systemic clearance or distribution may also be involved.

31.3 A diagnostic table

Observation	Plausible first hypotheses
Lower $C_{\max}$, similar AUC	slower absorption, delayed gastric emptying, extended release.
Lower $C_{\max}$, lower AUC, similar terminal slope	reduced extent of absorption or first-pass availability.
Same $C_{\max}$, lower AUC	faster elimination, altered late exposure, sampling issue.
Later $T_{\max}$, similar AUC	absorption delay without major extent loss.
Terminal slope after oral much slower than IV	possible flip-flop kinetics or depot input.

32 Expanded Atlas IV: Distribution, Binding, and Free Drug

32.1 Three concentrations

Many clinical arguments become clearer when we separate total plasma concentration, unbound plasma concentration, and tissue/effect-site concentration:

$$C_{\text{total}}, \qquad C_u = f_u C_{\text{total}}, \qquad C_{\text{site}}.$$

Assays often report total plasma concentration. Enzymes, transporters, and receptors may respond to unbound concentration. The effect site may not equilibrate instantly with plasma. A protein binding change can therefore alter interpretation even when total concentrations appear reassuring.

32.2 The free drug principle and its limits

The free drug principle says unbound drug is generally available for distribution, clearance, and receptor binding. But clinical inference needs caution. If binding decreases, total concentration may fall because unbound clearance increases, while unbound concentration may change little at steady state for some low-extraction drugs. For other drugs, unbound exposure changes meaningfully. The question is not whether free drug matters. The question is which quantity is controlled and which quantity is observed.

32.3 Distribution is often the hidden delay

Suppose a drug has rapid plasma decline but slow tissue equilibration. Early plasma samples may overstate elimination if distribution is not modeled. Conversely, a peripheral site of action may show delayed effect even when plasma concentration falls. The two-compartment model is the first mathematical language for this:

$$C_p(t) = Ae^{-\alpha t} + Be^{-\beta t}.$$

The early α phase often reflects distribution. The terminal β phase reflects the slowest coupled process observable in plasma. Clinical interpretation requires knowing which part of the curve is being used for which decision.

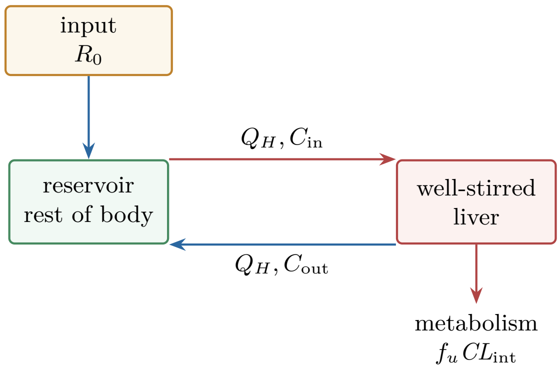
33 Expanded Atlas V: Hepatic Clearance and First Pass

33.1 The liver as a flow-capacity-binding system

The well-stirred model

$$CL_H = \frac{Q_H f_u CL_{\text{int}}}{Q_H + f_u CL_{\text{int}}}$$

compresses three controls: blood flow, unbound fraction, and intrinsic metabolic capacity. This one equation explains several clinical patterns. For high-extraction drugs, clearance is flow-limited and first-pass extraction can be high. For low-extraction drugs, clearance is more sensitive to $f_u CL_{\text{int}}$. Enzyme inhibition matters most when the inhibited pathway carries a substantial fraction of clearance.



$$CL_H = \frac{Q_H f_u CL_{\text{int}}}{Q_H + f_u CL_{\text{int}}}, \quad E_H = (C_{\text{in}} - C_{\text{out}})/C_{\text{in}}$$

Figure 10: Well-stirred hepatic clearance. This redraw keeps the source figure's core idea: blood flow brings drug into a mixed liver space where intrinsic capacity and binding determine extraction.

33.2 First pass and route dependence

For an oral drug,

$$F_H = 1 - E_H.$$

If E_H is high, oral bioavailability can be low even when absorption across the gut wall is complete. The same drug given IV bypasses first pass. Therefore oral and IV exposure comparisons can separate systemic clearance from bioavailability:

$$F = \frac{AUC_{\text{oral}}/Dose_{\text{oral}}}{AUC_{\text{IV}}/Dose_{\text{IV}}}.$$

This equation is simple but powerful because it compares dose-normalized exposure across routes.

33.3 DDI logic

For CYP3A-mediated interactions, the question is not only whether the perpetrator inhibits CYP3A. The question is how much of the victim drug's clearance or first-pass loss depends on CYP3A. A strong inhibitor can have small effect if f_m is small. A moderate inhibitor can have large effect if the pathway dominates. The same logic applies to induction, transporter inhibition, and combined enzyme-transporter interactions.

34 Expanded Atlas VI: Renal Clearance and Urine Data

34.1 Filtration, secretion, reabsorption

Renal clearance combines filtration, secretion, and reabsorption:

$$CL_R = f_u GFR + CL_{\text{secretion}} - CL_{\text{reabsorption}}.$$

If $CL_R \approx f_u GFR$, filtration is plausible. If $CL_R > f_u GFR$, active secretion is suggested. If $CL_R < f_u GFR$, reabsorption or incomplete urinary recovery is suggested. Urine pH matters mainly when passive reabsorption of ionizable drug is important.

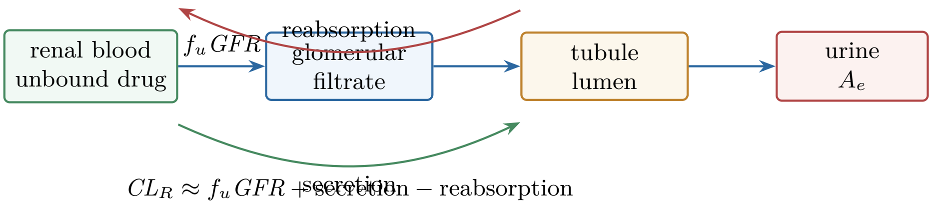


Figure 11: Renal clearance as filtration, secretion, and reabsorption. The drawing deliberately separates tubule lumen from blood so CL_R can be read mechanistically.

34.2 Amount excreted unchanged

The fraction excreted unchanged is

$$f_e = \frac{A_{e,\infty}}{Dose}.$$

For IV dosing under linear PK,

$$f_e = \frac{CL_R}{CL}.$$

This ratio helps decide whether renal impairment should affect exposure strongly. If renal clearance is a small fraction of total clearance, renal impairment may have limited parent-drug effect, though active metabolites or uremic transporter/enzyme changes may still matter.

34.3 Urine collection as an experiment

Urine data are only as good as the collection. Missing urine, wrong intervals, incomplete bladder emptying, and timing errors can distort renal clearance. The renal model includes patient behavior and nursing workflow. A beautifully fitted sigma-minus plot cannot repair a poor collection design.

35 Expanded Atlas VII: Dose Regimen Design

35.1 A regimen is a control policy

A regimen is not just a dose and interval. It is a control policy under uncertainty:

$$a = (Dose, \tau, \text{route}, \text{duration}, \text{monitoring rule}, \text{hold rule}).$$

The first design uses population expectations. Subsequent designs use observed response and concentrations. The regimen therefore evolves as information arrives.

35.2 Average, peak, trough, and fluctuation

For repeated dosing under linear PK,

$$C_{ss,\text{avg}} = \frac{F\,Dose}{CL\tau}.$$

Changing dose changes average exposure. Changing interval changes average exposure and fluctuation. For the same daily dose, shorter intervals usually reduce peak-trough swing; longer intervals increase swing. Whether this matters depends on the therapeutic target and toxicity mechanism.

35.3 A practical design sequence

One disciplined sequence is:

(i) choose target exposure or target concentration range;

(ii) estimate CL for maintenance;

(iii) estimate V for loading if rapid target achievement is needed;

(iv) choose interval from half-life, toxicity, convenience, and fluctuation tolerance;

(v) simulate peak, trough, and AUC;

(vi) plan when to measure and how to update.

This sequence is more robust than memorizing dose tables because it says why each part of the regimen exists.

36 Expanded Atlas VIII: Nonlinear Dose Adjustment

36.1 Why linear intuition breaks

Linear PK gives

$$AUC \propto Dose.$$

Nonlinear PK breaks this proportionality. Near capacity limitation, the next dose increase can produce a disproportionate exposure increase. This is why nonlinear drugs demand humility. A patient can be stable at one dose and toxic after a small increase if the system is close to saturation.

36.2 The phenytoin denominator

At steady state,

$$C_{ss} = \frac{K_m R}{V_{\max} - R}.$$

The denominator $V_{\max} - R$ is the clinical warning. If R approaches $V_{\max}$, concentration can rise steeply. The equation also explains why time to steady state is not fixed: apparent half-life lengthens as concentration approaches saturating ranges.

36.3 Nonlinearity versus time dependence

Some PK is nonlinear because rates depend on concentration. Some is time dependent because the system changes: autoinduction, autoinhibition, enzyme turnover, disease progression, or antibody formation. The observed data may show changing clearance over time. The model should not force all deviations into a concentration-dependent $CL(C)$ if biology suggests time-varying capacity.

37 Expanded Atlas IX: Special Populations as Mechanistic Perturbations

37.1 Do not adjust by labels alone

Special populations are not magic categories. They are perturbations of mechanisms. Pediatrics perturbs size and maturation. Pregnancy perturbs volume, GFR, enzymes, transporters, plasma proteins, and blood flow. Obesity perturbs body composition and sometimes clearance. Renal disease perturbs filtration, secretion, binding, metabolites, and nonrenal pathways. Hepatic impairment perturbs flow, enzymes, transporters, shunting, bile, and synthesis.

37.2 The dose question changes by parameter

If the main change is V, loading dose changes. If the main change is CL, maintenance dose changes. If F changes, oral and IV regimens diverge. If PD sensitivity changes, target exposure itself may change. A special-population dose recommendation therefore requires specifying which layer changed:

$$V, \quad CL, \quad F, \quad \text{target}, \quad \text{safety margin}, \quad \text{monitoring uncertainty}.$$

37.3 Clinical covariates as imperfect measurements

Creatinine clearance, eGFR, Child–Pugh class, albumin, bilirubin, body weight, gestational age, and genotype are imperfect measurements of underlying physiology. A covariate model is useful not because the covariate is the mechanism itself, but because it carries information about the mechanism. Population modeling formalizes this imperfect translation.

38 Expanded Atlas X: Biologics and Modality-Specific PK

38.1 Antibodies

Antibodies often have long half-lives because FcRn recycling protects them from degradation. Their distribution is limited by size, convection, binding, and tissue penetration. Their clearance can be linear through nonspecific catabolism and nonlinear through target-mediated pathways. The same antibody can show concentration-dependent clearance at low concentrations and approximately linear clearance at high concentrations.

38.2 ADCs

For ADCs, the analyte matters. Total antibody, conjugated antibody, antibody-conjugated payload, free payload, and metabolites can all tell different stories. A stable linker may reduce premature payload release but may also affect intracellular release. A high drug-antibody ratio may increase potency but alter clearance or aggregation. The PK problem is inseparable from chemistry and cell biology.

38.3 Oligonucleotides

Oligonucleotides have absorption, distribution, and PD features unlike classical small molecules. They can have rapid plasma decline but long tissue half-life. The relevant exposure may be tissue concentration or target RNA knockdown rather than plasma AUC. Chemical modification, GalNAc targeting, renal clearance, protein binding, and endosomal escape all become part of the PK/PD chain.

38.4 CAR-T and living drugs

CAR-T therapy makes the word pharmacokinetics metaphorical but still useful. The administered cells expand, traffic, kill, contract, and sometimes persist. Exposure may be cell count over time. Response depends on antigen, tumor burden, immune state, cytokines, and exhaustion. Dose starts a dynamical system rather than filling a compartment with passive molecules.

39 Expanded Atlas XI: Building a Population Model

39.1 The structural model

Start with the forward map:

$$\hat{y}_{ij} = g(t_{ij}, u_i, \psi_i).$$

This could be a closed-form one-compartment equation, an ODE solution, a TMDD system, or a PBPK model. The structural model should be driven by the question and data richness. Sparse trough-only data cannot support the same structural ambition as dense Phase 1 profiles.

39.2 The statistical model

Next define variability:

$$\psi_i = h(c_i, \theta, \eta_i), \qquad \eta_i \sim \mathcal{N}(0, \Omega).$$

Then define residual error:

$$y_{ij} = g(t_{ij}, u_i, \psi_i) + \epsilon_{ij} \quad \text{or} \quad \log y_{ij} = \log g(t_{ij}, u_i, \psi_i) + \epsilon_{ij}.$$

Interindividual variability and residual variability answer different questions. A high Ω says patients differ. A high Σ says observations mismatch predictions after accounting for individual parameters.

39.3 Covariate modeling

A covariate should be clinically plausible, statistically visible, and decision relevant. Weight on volume is plausible; renal function on renal clearance is plausible; genotype on metabolic clearance may be plausible. But the point is not to decorate the model with significant predictors. The point is to explain variability in a way that improves prediction or decisions.

39.4 Evaluation

Model evaluation should include goodness-of-fit plots, residual diagnostics, prediction-corrected VPC where needed, bootstrap or sampling uncertainty, shrinkage assessment, sensitivity to structural alternatives, and checks against the decision. A model can fit observed concentrations and still be poor for dose selection if it predicts future troughs badly.

40 Expanded Atlas XII: NONMEM Reading Notes

40.1 Reading control streams

When reading NONMEM code, translate each block:

$INPUT	:	data columns and meaning,
$PK	:	individual parameter construction,
$DES	:	ODE system, if present,
$ERROR	:	observation and residual error model,
$THETA	:	typical values and covariate effects,
$OMEGA	:	random effects variance,
$SIGMA	:	residual variance.

This translation turns code into the hierarchy $p(y_i \mid \psi_i)p(\psi_i \mid c_i)$.

40.2 Shrinkage

Empirical Bayes estimates can shrink toward the population mean when individual data are sparse or noisy. Shrinkage is not inherently bad; it is a property of borrowing strength. But high shrinkage makes individual ETA-based diagnostics and covariate plots less informative. A model report should state when individual estimates are data-driven and when they are mostly population-driven.

40.3 VPC and PPC

Visual predictive checks compare observed data to simulated data from the fitted population model. They ask whether the model can reproduce central tendency and variability over time, dose, covariate, or bin. A posterior predictive check is the Bayesian cousin. Both are simulation-based humility: if the fitted model cannot simulate data like what was observed, the likelihood and parameters are not enough.

41 Expanded Atlas XIII: PBPK Credibility

41.1 System and drug parameters

PBPK separates system parameters from drug parameters. System parameters include tissue volumes, blood flows, enzyme abundance, transporter expression, GFR, hematocrit, and demographics. Drug parameters include partition coefficients, permeability, intrinsic clearance, solubility, binding, transporter kinetics, and degradation. This separation enables virtual populations and extrapolation, but only if uncertainty is tracked.

41.2 Sensitivity

For a PBPK model, ask which parameters control the prediction:

$$S_k = \frac{\partial \log y}{\partial \log \theta_k}.$$

If the decision depends on a parameter that is poorly measured and highly sensitive, the model needs either more evidence or more cautious claims. If a parameter is insensitive for the question, elaborate measurement may not be worth the effort.

41.3 Credibility statement

A PBPK report should state:

(i) question of interest;

(ii) context of use;

(iii) model structure;

(iv) data used for parameterization;

(v) verification checks;

(vi) validation or qualification cases;

(vii) uncertainty and sensitivity;

(viii) decision consequence.

This is as much scientific writing as computation.

42 Expanded Atlas XIV: PD, Delays, and Turnover

42.1 Choosing among direct, effect-site, and turnover models

If response follows concentration immediately and reversibly, direct effect may be enough. If response lags because biophase concentration equilibrates slowly, use an effect compartment. If response lags because the biomarker is produced and lost over time, use an indirect response model. If response depends on cells, disease burden, immune feedback, or tolerance, a larger mechanism may be required.

42.2 Hysteresis

Plot effect versus concentration over time. Counterclockwise hysteresis often suggests delayed effect-site equilibration or response turnover. Clockwise hysteresis can suggest tolerance, feedback, depletion, or time-dependent sensitivity. Hysteresis is a visual symptom; the mechanism still needs pharmacological reasoning.

42.3 Indirect response time scale

For

$$\frac{\mathrm{d}R}{\mathrm{d}t} = k_{\mathrm{in}} - k_{\mathrm{out}} R,$$

baseline is $R_0 = k_{\mathrm{in}}/k_{\mathrm{out}}$, and response half-life is $\log 2/k_{\mathrm{out}}$. If a drug inhibits production, response declines according to turnover. This explains why drug concentration can change quickly while biomarker response changes slowly.

43 Expanded Atlas XV: QSP as Organized Ambition

43.1 When to escalate

Escalate from PK/PD to QSP when the scientific question requires interacting mechanisms: immune activation, tumor-immune dynamics, cytokine feedback, receptor trafficking, disease progression, combination therapy, resistance, or systems-level biomarkers. Do not escalate merely because a model is impressive. Complexity should buy decision-relevant structure.

43.2 State, parameter, observation, decision

Every QSP model should still be expressible as

$$\frac{\mathrm{d}x}{\mathrm{d}t} = f(x, u, \theta, t), \qquad y = g(x, \theta, t) + \epsilon, \qquad a = \delta(y, c).$$

The state equation is not enough. Observations and decisions complete the model. A QSP model that cannot say what is observed and what decision it informs is a mechanistic diagram, not yet a clinical pharmacology tool.

43.3 Model reduction

Large systems often contain directions that barely affect the prediction of interest. Sensitivity analysis, sloppiness, MBAM-like thinking, and modular reduction ask which mechanisms are essential for the question. Reduction is not anti-mechanistic. It is the discipline of retaining the mechanisms that matter for the decision.

44 Expanded Atlas XVI: Study Problems

44.1 Core problems

(i) A drug has IV $AUC = 40\ mg\,h/L$ after 200 mg. Estimate CL. If target $C_{ss,\text{avg}}$ is 5 mg/L, what infusion rate is suggested?

(ii) A 500 mg oral dose gives $AUC = 25\ mg\,h/L$. A 250 mg IV dose gives $AUC = 50\ mg\,h/L$. Estimate absolute bioavailability.

(iii) A patient's trough is high but peak is acceptable. Which regimen changes reduce trough without increasing peak?

(iv) A drug's terminal slope after oral extended-release dosing is slower than after IV dosing. Explain two possible mechanisms.

(v) For a low-extraction drug, predict the effect of doubling f_u on total clearance, total concentration, and unbound concentration at steady state. State assumptions.

44.2 Modeling problems

(i) Write a one-compartment oral model with first-order absorption and first-order elimination. Identify state variables, parameters, input, and output.

(ii) Extend it to include an effect compartment. What new parameter appears, and what clinical pattern does it explain?

(iii) Write a lognormal population model for CL with weight and creatinine clearance covariates.

(iv) Explain why sparse trough data may identify CL better than V.

(v) Design a VPC for a multiple-dose study with unequal sampling times.

44.3 Mechanistic problems

(i) A monoclonal antibody shows nonlinear clearance at low dose but linear clearance at high dose. Sketch a TMDD explanation.

(ii) An ADC has discordant total antibody and free payload profiles. What mechanisms could explain the mismatch?

(iii) A biomarker response continues after plasma drug is gone. Compare active metabolite, effect-site delay, and indirect response explanations.

(iv) A PBPK model predicts a strong DDI driven by hepatic uptake transporter inhibition. What evidence would increase credibility?

(v) A QSP model has 80 parameters. What would you inspect before trusting it for dose selection?

45 Appendix: Formula Map

45.1 Core PK

$$
\begin{aligned}
C(t) &= C_0 e^{-k_e t}, \\
t_{1/2} &= \frac{\log 2}{k_e}, \\
CL &= k_e V, \\
AUC &= \frac{F\,Dose}{CL}, \\
C_{ss,\mathrm{avg}} &= \frac{F\,Dose}{CL\tau}, \\
LD &= \frac{C_{\mathrm{target}}\,V}{F}, \\
MD/\tau &= \frac{C_{\mathrm{target}}\,CL}{F}.
\end{aligned}
$$

45.2 Absorption and infusion

$$
\begin{aligned}
C_{\mathrm{oral}}(t) &= \frac{F\,Dose\,k_a}{V(k_a - k_e)} \left(e^{-k_e t} - e^{-k_a t}\right), \\
C_{\mathrm{inf}}(t) &= \frac{R_0}{CL}(1 - e^{-k_e t}), \\
C_{ss} &= \frac{R_0}{CL}.
\end{aligned}
$$

45.3 Organ clearance

$$
\begin{aligned}
CL_{\mathrm{organ}} &= QE, \\
CL_H &= \frac{Q_H f_u CL_{\mathrm{int}}}{Q_H + f_u CL_{\mathrm{int}}}, \\
F &= F_a F_G F_H.
\end{aligned}
$$

45.4 Nonlinear and PD

$$
\begin{aligned}
\frac{\mathrm{d}A}{\mathrm{d}t} &= -\frac{V_{\max} C}{K_m + C}, \\
E(C) &= E_0 + \frac{E_{\max} C}{EC_{50} + C}, \\
\frac{\mathrm{d}C_e}{\mathrm{d}t} &= k_{e0}(C_p - C_e), \\
\frac{\mathrm{d}R}{\mathrm{d}t} &= k_{\mathrm{in}} S(C) - k_{\mathrm{out}} R.
\end{aligned}
$$

45.5 Population model

$$
\begin{aligned}
\psi_i &= h(c_i, \theta, \eta_i), \qquad \eta_i \sim \mathcal{N}(0, \Omega), \\
y_{ij} &= g(t_{ij}, u_i, \psi_i) + \epsilon_{ij}, \\
p(\psi_i \mid y_i, c_i, u_i, \theta) &\propto p(y_i \mid \psi_i, u_i) p(\psi_i \mid c_i, \theta).
\end{aligned}
$$

46 Closing Synthesis

The three-course sequence is one intellectual arc. PHC 607 teaches the grammar of concentration-time data. PHC 608 teaches how that grammar becomes dosing. PHC 609 teaches how dosing becomes inference under biological variability, nonlinear mechanisms, sparse data, delayed response, and population-level learning.

The deepest unification is this:

$$\text{clinical pharmacology} = \text{mechanistic forward model} + \text{statistical observation model} + \text{decision under uncertain}$$

Once this sentence is internalized, the topics stop feeling like a catalog. Clearance, volume, bioavailability, NCA, compartment models, TDM, special populations, biologics, TMDD, PBPK, PopPK, PD, and QSP become different ways of answering the same humane question: given what we know about this drug, this patient, this disease, and this evidence, what should we do next?

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